CARPET: Coarse Action Reinforcement Policy Extraction Tool

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We start by creating a partitioning based on a select set of predicates over the state space. Then we use a dataset D of observations of state-action pairs from n trajectories of the environment[1] to split those regions where the agent will either choose action 0 ('no hit') or action 1 ('hit'). This is done using polynomial logistic regression with each observed state being labelled according to the action associated with it. The degree of the polynomial is chosen by starting with simple linear regression (ie. degree 1) and repeating the regression with an incremented degree if the accuracy of the split is lower than 0.95.

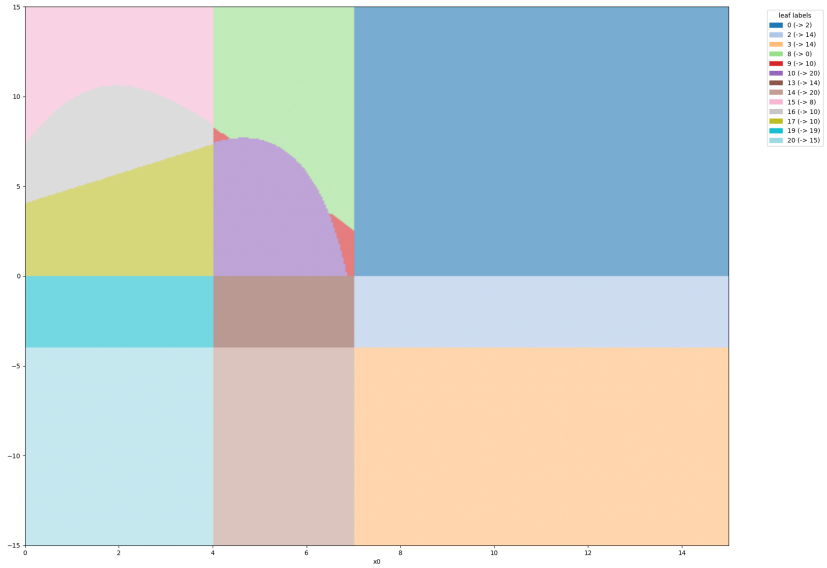


Figure 1: State space partitioning after splitting according to action. The axis-aligned splits are based on the initial predicates, the other are based on polynomial regression to separate regions based on the action performed. Regions in this figure are not labelled with their corresponding action, but uniquely colored. The legends shows what the most likely next region is from each region.

Figure 1 shows the emerging regions in different colors. Furthermore, we

[1] In this setup $n = 100$ and the well-trained agent reaches 400 timesteps in each run resulting in $|D| = 400000$

use D to estimate the next-region transitions for each region. That is, for each region ν_i we observe a set of states $S_{\nu_i} \subseteq D$ that fall within that region. Let $s_{j,t} \in S_\nu$ be the state we encountered in the jth trajectory at timestep t. For each state $s_{j,t} \in S_{\nu_i}$ we check what region $s_{j,t+1}$ belongs to and a get a next-region transition probability estimate based on D.

Assessing the quality of this model is done by simply executing the environment again and now comparing the predicted next regions with the actual next region. In that case, this initial version as an accuracy of 0.69 (for action prediction the accuracy is 0.99).

Now we try to refine the model by splitting regions according to the next-region transitions of its states using a fresh dataset D. If the states in S_{ν_1} transitions to more than 1 other state, we first search for a split that separates states going to the most common next-region from all the rest and then we repeat recursively in the new regions.[2] By that process, we go from 21 to 37 regions and we increase the accuracy to 0.73.

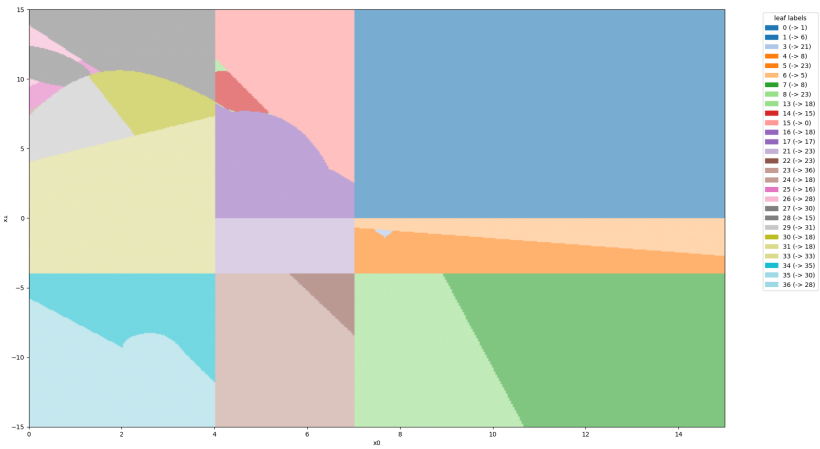


Figure 2: The partitioning after the first round of refinement. Note that the color map cannot represent all the regions with distinct colors, making the interpretation less obvious.

We can repeat this process to get a higher precision but increasing the number of regions even more. In Figure 3 we see the partitioning after the second round of next-region based splits, where the partitioning contains 71 regions and incurs an accuracy of 0.77. However, the splits are becoming increasingly complicated (that is, more regions are split according to high-degree polynomial) and for several regions the split is based on very few data points. Repeating the splitting process after this second round does not yield any noticable improvements in accuracy, but amplifies the aforementioned problems.

Another problem occurs in the regions where the next-region transition is stochastic. For the very coarse original partitioning (in this particular example), we observed no stochasticity in the transitions. But as the partitioning is

[2] As I am writing this, I am thinking, that maybe we should start by separating the states that go from region ν to ν (ie. stay within the same region), so that we don't mess up the structure?

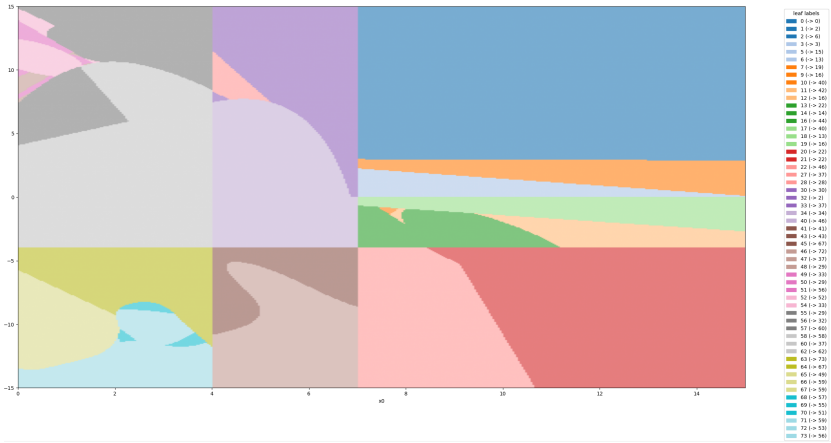


Figure 3: The partitioning after the second round of refinement. The legend is now no longer able to show all the regions.

refined, regions appear where states can transition to mulitple different regions dependent only on unknown randomness, thus making a clean split impossible.

In Figure 4 we show an example of this. The colored stripes represent regions where as the small crosses represent observed states. The color of the crosses indicate what region that state transitions to at the next state. Note that these states and transitions are sampled from actual executions of the environment and the transitions are not deterministic. We do see some pattern in the transitions, but it is also clear that it is impossible to do a perfect split. Thus, we need to come up with a different approach.

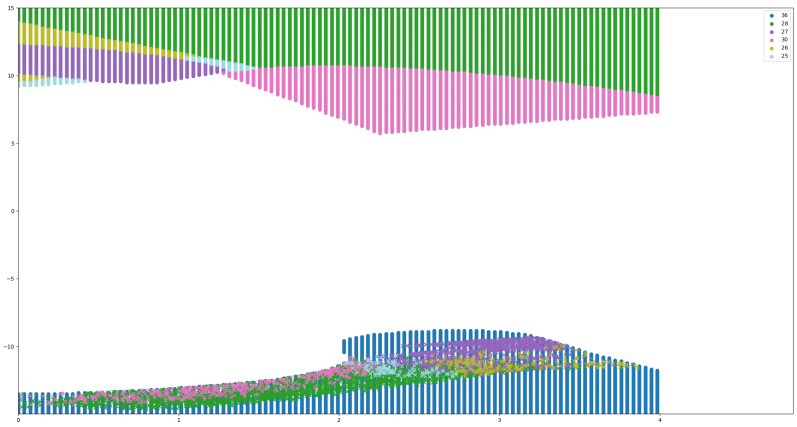


Figure 4: A closer look at the states (the colored crosses) in a given region (the blue striped area) and the regions that the states transitions to in our sampling data (the color of a cross correspond to the color of the striped region it will transition to).

One approach that we have discussed earlier is to characterize each state by the probability distribution of its next-region transition. This would require that

we take each observed state (or we sample states systematically or randomly) and then sample adequately many next-states to get a sample distribution of the next-region transition probability. We would then need to learn a mapping from a state to a probability distribution and then do our split according to this.

0.1 Checking for next actions

These next images show some instances where the next regions of a state is a 'hit'-state. Here, my idea have been to make splits according to the future actions, so in that way predict the future behavior. I have not yet implemented splits on this condition, but nonetheless, it might be interesting.

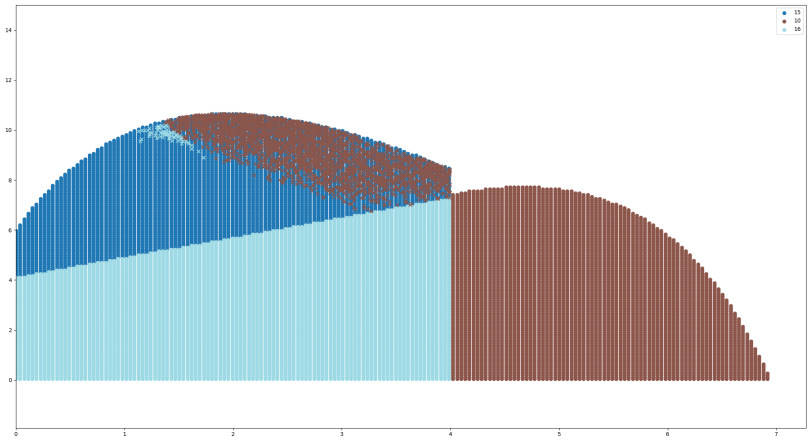


Figure 5: In the blue region, the brown crosses will go to the brown state, where the action is 'hit'. The light-blue crosses will go to the light-blue state.

1 Clustering and Total Variational Distance

In this section, we look at using agglomorative clustering to come up with region splits. We also use Total Variational Distance (TV) to assess the quality of a split. As the splitting procedure begins, the TV distance is at 1 every time. When we encounter lower values, we inspect that further (see below). First, we take a look at some unwanted behavior earlier.

In Figure 6 we see a region whose points are separated into 2 clusters (red and pink), which is to be used as the basis for splitting the region in 2. While the pink cluster seems fairly coherent, the red points are scattered in 2 different locations. One that seem to have a natural separation border with the pink cluster, and another along the border of the region itself, lumped in with the pink points.

Upon further inspection, we see that the pink cluster is deterministic in its transition probability, that is, all points transition to the same next region (region 10). The red cluster, however, transitions to 3 different regions. The

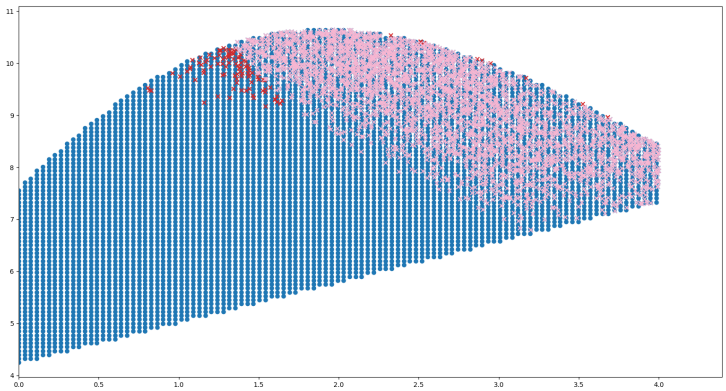


Figure 6: All the pink crosses transitions to the same region. The red ones have 3 different possible regions to go to. Especially, notice the ones in the top 'right' following the edge of the region boundary.

points spread along region border (together with the pink points) all transition to region 8, while the rest go to either region 17 (predominantly) or 16.

The **question becomes**: how can we make a separation that makes the new regions convex? In this example, the few red points among the pink ones would be better to include in the pink cluster even though they 'pollute' the transition probaility a little bit.

Another example of conflicting clusters within a region is given in Figure 7. This is a result of the decision boundary created in an earlier iteration not being perfect — which is impossible to achieve. This region should therefore not be split anymore. But, **how do we recognise this situation and choose not to split?**

Low td value In the Figure 8, we also see a non-convex cluster (the red one) but furthermore, we see that the Total Variational Distance between the two clusters (measured as the TV distance between the probability vectors associated with the points in the cluster) is lower than 1.

By closer inspection it is also seen, that within both groups some states have a most likely transition to state 16. In previous splits, no such overlap have happened. This is most likely because the transitions from this region has a stochastic component — how much 'bounce' happens when the ball hits the ground (position gets to zero and velocity goes from negative to positive).

We can zoom in to find that the red points with the higher x-value can actually be perfectly separated from the pink points. However, the other part of the red cluster overlaps with the pink cluster, see Figure 9.

By letting the clustering algorithm find 3 clusters instead of 2, we get 2 clusters, call them a and b, that each has a single 'most likely' transition (to region 16 and 20, respectively). The third, call it c, has states that most likely transitions to region 15 and others that transition to region 16. The TD value between a and b are 0.99, and so is it between b and c which don't share any most likely next region. However, between a and c the TD value is only 0.77. Figure 10 shows this separation.

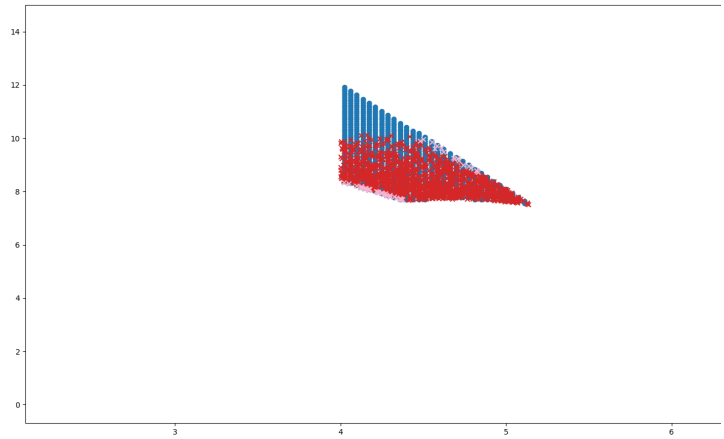


Figure 7: In this example, the red points represent the intended points for this region, ie. this region is supposed to describe an area, where all states transition to the region that the red points go to. The pink points are a result of inevitable misclassification during the last split and actually belong to neighbouring regions. Therefore, this region should not be split any further.

Now we try and exploit this, by making our splitting procedure check the TD distance before settling in on a split. If it is lower than 0.95, we try and increase the number of clusters. Then we look for the cluster with the largest TD distance to the other clusters, and if this is large enough, we split the region between that cluster and all the rest. For the region we looked at in Figure 8, that results in the wanted split, where red points with the larger x-value are confined to their own region.

However, when the procedure repeats and gets back to remainder of the original region where the clusters were overlapping, we get the result seen in Figure 11 when we make 3 clusters. Here, all 3 clusters have an overlap with one another.

In Figure 12 we show the same region but where the points have been labelled according to wether their transition probability vector assigns all probability to a single transition (the red crosses) or spreads it accros multiple regions (the pink crosses). We see 3 distinct areas, where the (sampled) transitions are deterministic and a greater middle area with stochastic transitions.
Ideas for how to separate these points (in Figure 12):

- Create 3 new regions, each characterized by the most probable transition. This would mean that each region is formed around the red points but contain the pink points whose transition probabilities are closest to those of the corresponding red points.

- Create 3 new regions that (almost) *only* contain the red points and then another region containing the pink points; this would result in 3 regions with a very high certainty about the next region and then one region with a low certainty.

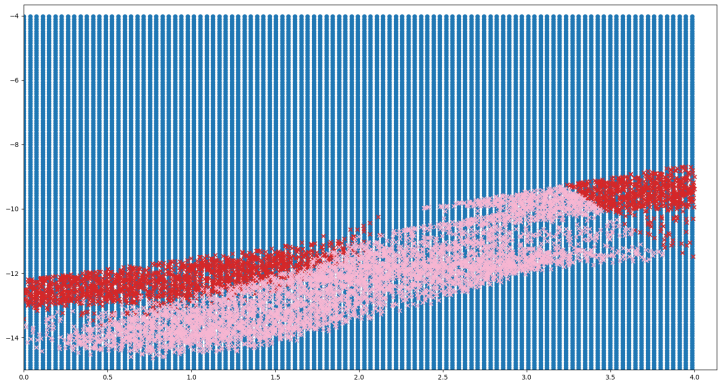


Figure 8: separation-low-td-val

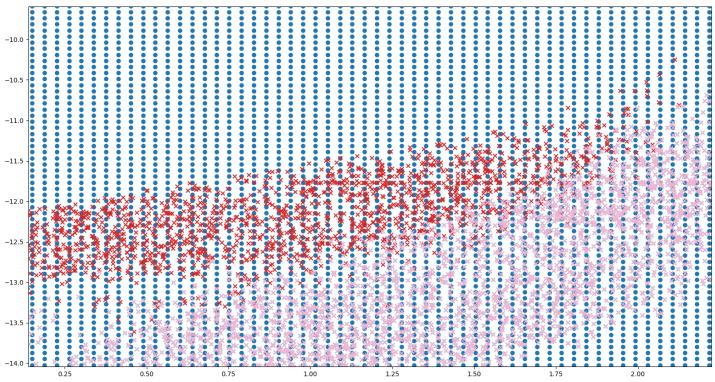


Figure 9: separation-low-td-val-zoom

2 Deterministic transitions

How do we separate regions where all states have deterministic transitions, but there are 3 or more 'groups' of transitions (ie. a group of states going to region x, another to region y and yet another to region z)? One idea is simply to separate one group from all the other. However, if we look at an example such as the one in Figure 13, we see that the choice of the group to split from the rest will have a pretty significant impact on what kind of split can be made.

Whatever single group in Figure 13 we choose to split from the rest, we would not be able to make an accurate split with a simple, low-degree polynomial. However, it looks like we could separate the purple and yellow groups from the green and pink groups by a nice linear boundary. As such, the naive approach seems flawed. But how do we identify the good split? We could test out all possible splits and select the one that has the best score balanced with the lowest degree.

Another example of this is given in Figure 14a and 14b, where we both see the discrete groups and the way they are split when using aggloagglomorative clustering to separate 1 from the other.

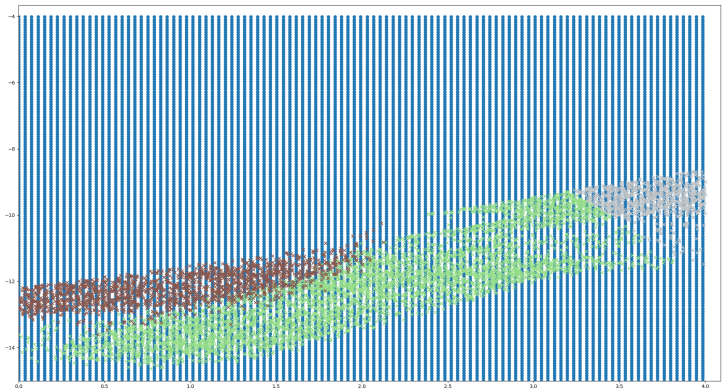


Figure 10: separation-low-td-3-clusters

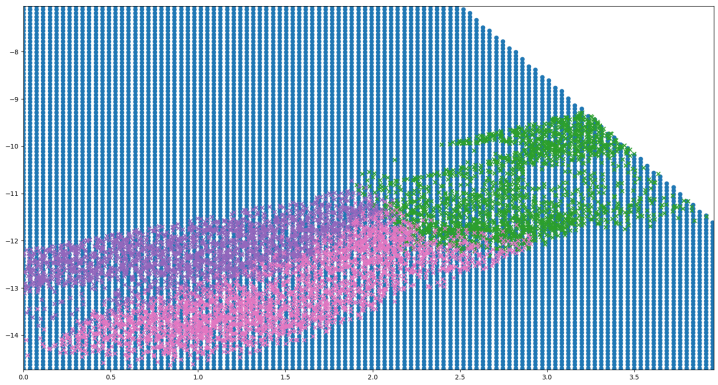


Figure 11: These 3 clusters are not perfectly separated. How to choose an appropriate split?

In the next example, Figure 15, we have a more difficult situation, that harkens back to Figure 7. Here we have 3 groups of deterministic transitions. However, one group (the grey one) is actually just a result of misclassification. That is, these points 'should' belong to a different region, but since our splits inevitably are not 100% accurate, we get these groups that should actually not be considering when splitting. We want to ignore such misclassified points and just consider them to belong to the group that they are closest to in *point-space* (in this case, the grey should belong to the green group). But how do we identify this?

2.0.1 Problem?

Imagine we have a 1-dimensional enviromnent, where a ball rolls along the x-axis. At each timestep it will move a distance of 1 plus some gaussian noise centered around 0. Can we partition this state space in such a way that it meaningfully captures the behavior of the system?

Let the dynamics of the system be such, that a ball observed in state s^t will

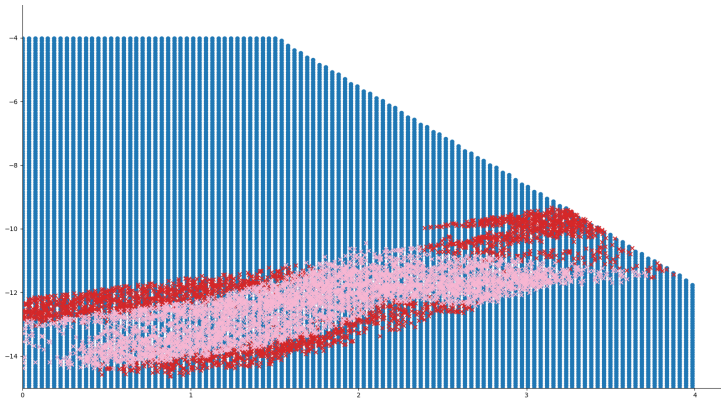


Figure 12: The red crosses are points with a probability vector assigning all probability on one transition. The pink crosses spreads the probability on multiple transitions.

move a distance of $d \in [1-a, 1+a]$, where a is some small positive constant. We define a starting region $r^0 = (r^0_{\min}, r^0_{\max})$ such that $r^0_{\max} - r^0_{\min} = 1 - a$ which ensures that whatever our initial state $s^0 \in r^0$, we are guaranteed that s^1 will have left r^0.

We now know that s^1 will end up somewhere between $r^0_{\min} + 1 - a$ and $r^0_{\max} + 1 + a$. If we use this to define the next region r^1, then we can guarantee that for any $s^0 \in r^0$ we have that $s^1 \in r^1$. But we can't do the same trick for r^2, since the domain of s^2 will be the range $[r^0_{\min} + 2 - 2a, r^0_{\max} + 2 + 2a]$. Since we defined $r^0_{\max} - r^0_{\min} = 1 - a$ and $a > 0$, we can see that

$$\begin{aligned} r^0_{\min} + 2 - 2a &< r^0_{\max} + 1 + a \\ 2 - 2a &< r^0_{\max} - r^0_{\min} + 1 + a \\ 2 - 2a &< 1 - a + 1 + a \\ 2 - 2a &< 2 \end{aligned}$$

This means, that s^2 *might* end up in r^1. Conversely, the upper bound of the domain of s^2 is obviously larger than the upper bound of r^1 as we have $r^0_{\max} + 2 + 2a > r^0_{\max} + 1 + a$, meaning that s^2 might also leave r^1. Therefore, we cannot for certain say what region states from r^1 will transition to — and the only way to achieve this is by splitting r^0 into smaller regions thereby losing our guarentee of the transition function from that region.

What if we define 2 sets of regions, R_i and R_j, such that for all regions $r^i \in R_i$ we have $r^i_{\max} - r^i_{\min} = 1 - a$, and for all regions $r^j \in R_j$ we have $r^j_{\max} - r^j_{\min} = (r^i_{\max} + 1 + a) - (r^i_{\min} + 1 - a) = 1 + a$?[3] Then for any region r^i, we know for certain that the transitions will go to $r^{i+1} \in R_j$, and for any r^j, we know that either s^{t+1} stays in r^j or s^{t+1} transitions to a region $r^{j+1} \in R_i$. More precisely, we know that in the range $[r^j_{\min}, r^j_{\min} + 2a]$ a state in r^j *may*

[3]Note, in the following we will sometimes just write r^j as a shorthand for 'a region $r \in R_j$'

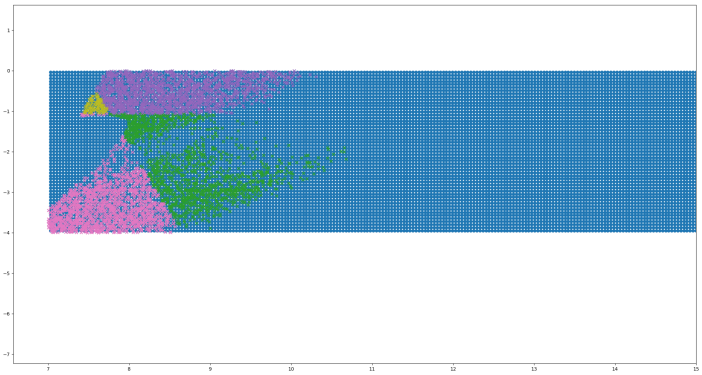


Figure 13: Each color crosses represent a unique deterministic transition. As such, there are 4 groups of points in this region, each group transitioning with 100% certainty to a specific next region.

stay in $r^j \in R_j$ and in the range $[r^j_{\min} + 2a, r^j_{\max}]$ it will *definitely* transition to $r^{j+1} \in R_i$.

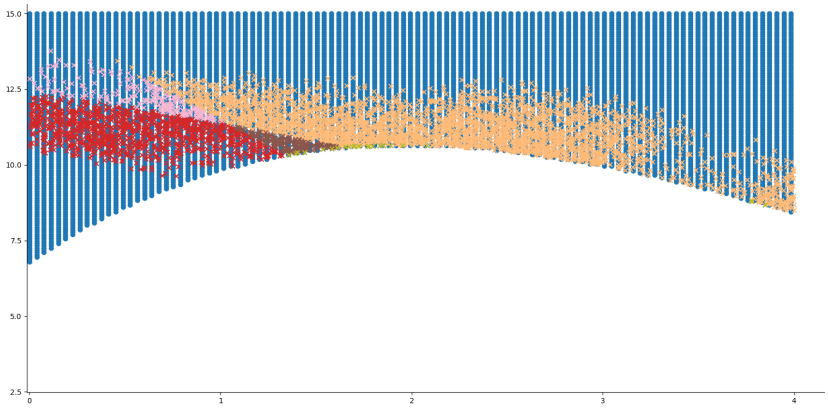
From this observation, consider the following: say we observe s^t in $r^j \in R_j$. Then there are 2 possible regions for s^{t+1}, either $r^j \in R_j$ or $r^{j+1} \in R_i$. However, given that we observed s^t in r^j, we can now say with absolute certainty what the region of s^{t+2} will be: if $s^{t+1} \in r^j$ then $s^{t+2} \in r^{j+1}$, and if $s^{t+1} \in r^{j+1}$ then $s^{t+2} \in r^{j+2}$, where $r^{j+2} \in R_j$. In other words, if the only thing we know about a state s is that at time t it was in $r^j \in R_j$ and we know what region it was in at time $t+1$, then we know excactly what region it will be in at time $t+2$.

But what if s^t was in $r^i \in R_i$? Then we know that s^{t+1} is in $r^{i+1} \in R_j$ and then we just need to know the region at time $t+2$ to determine (with certainty) the region at time $t+3$. This means, that by observing 3 regions in a row (starting from a region in R_i), we can determine the next region with absolute certainty.

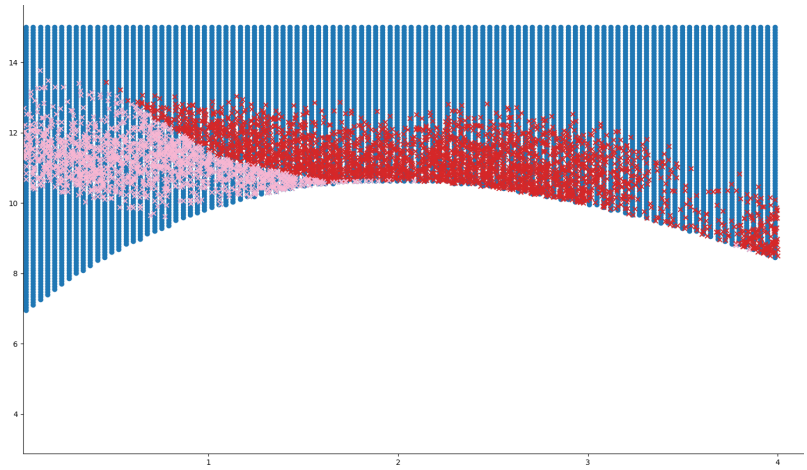
So, for a sequence of regions starting in R_j, we need just a sequence length of 2 to determine the next region. If the sequence is starting in R_i, we need 3. But, if we observe (r^j, r^i) then we might be able to determine that the next region is r^j, but now we cannot know what will follow that, since we are now stuck with a sequence of length 2 starting with r^i.

The possible region sequences are the following:

$$
\begin{gathered}
(r^i, r^j, r^i, r^j) \\
(r^i, r^j, r^j, r^i) \\
(r^j, r^i, r^j, r^i) \\
(r^j, r^i, r^j, r^j) \\
(r^j, r^j, r^i, r^j)
\end{gathered}
$$



(a) Deterministic groups



(b) 2 splits

Figure 14: Top: 5 distinct groups of deterministic transitions (yellow, pink, brown, red and green). Bottom: the split when just organising in 2 clusters.

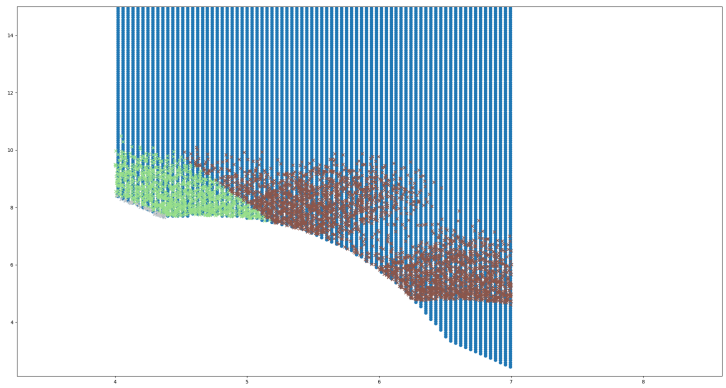


Figure 15: fig:deterministic-groups-misclassification1